



**A STUDY ON ARTIFICIAL INTELLIGENCE
ON EMPLOYEE PERFORMANCE AND
HUMAN RESOURCE IN SETNEXT
PVT. LTD ERODE.**



PROJECT REPORT

Submitted by

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731223631031

In Partial Fulfillment for The Award of The Degree

Of

MASTER OF BUSINESS ADMINISTRATION

In

DEPARTMENT OF MANAGEMENT STUDIES

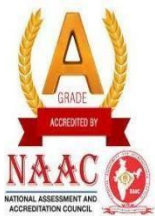
J.K.K.MUNIRAJAH COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

Accredited by NAAC with “A” grade

T.N. PALAYAM - 638 506.

MAY 2025



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**DEPARTMENT OF MANAGEMENT STUDIES
PROJECT REPORT**

MAY 2025

**This is to certify that the project entitled
A STUDY ON ARTIFICIAL INTELLIGENCE ON EMPLOYEE
PERFORMANCE AND HUMAN RESOURCE IN
SETNEXT PVT. LTD, ERODE.**

is the Bonafide record of project work done by

MOHAMMED SANOOB

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Of MBA during the year 2023 to 2025.

Project Guide

Head of the Department

Submitted for the Project Viva-Voce examination held on _____

Internal Examiner

External Examiner

DECLARATION

I affirm that the project work titled **(A STUDY ON ARTIFICIAL INTELLIGENCE ON EMPLOYEE PERFORMANCE AND HUMAN RESOURCE IN SETNEXT PVT. LTD, ERODE.)** being submitted in partial fulfillment for the award of Master of Business Administration is the original work carried out by me. It has not formed the part of any other project work submitted for award of any degree or diploma, either in this or any other University.

MOHAMMED SANOOB

731223631031

I certify that the declaration made above by the candidate is true.

Signature of the Guide,

Mr.R.PRASANTH.MBA.,

(Asst Professor)

ACKNOWLEDGEMENT

I extend my deep sense of gratitude and sincere thanks to our Chairman **Mrs.VASANTHAKUMARI MUNIRAJAH -J.K.K.MUNIRAJAH** group of Institutions For giving me an opportunity to be a student of this reputed institution.

I extend my deep sense of gratitude and sincere thanks to our Secretary **Mrs.KASTHURIPRIYA .MBA -J.K.K.MUNIRAJAH** group of Institutions for giving Me an opportunity to be a student of this reputed institution.

I extend my respect to our Principal **Dr.K.SRIDHARAN., M.E., M.B.A., Ph.D.,** For his valuable support in carrying my project work.

It is my privilege to thank our Head of the Department Asst.Prof. **E.R.VADIVEL, M.B.A.,** Department of Management Studies for his support and guidance for doing my project work.

I express my sincere thanks to my project guide **Mr. R. PRASANTH.MBA.,** for his valuable suggestion and guidance for doing the project work.

I also express my sense of gratitude to **Mr. SENTHIL KUMAR** for providing me this wonderful opportunity to do my project work in the topic, **“A STUDYARTIFICIAL INTELLIGENCE ON EMPLOYEE PERFORMANCE AND HUMAN RESOURCE IN SETNEXT PVT LTD”.**

I also express my gratitude to all the faculty members, parents and my fellow mates who have helped me to carry out this work. Last but not the least; I thank my almighty God for His blessing showed on me during this period.

MOHAMMED SANOOB

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INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **RAGUL RAJ** has successfully completed his internship as a **Software Developer** at **KODEX INFOSOLUTIONS** during the period from **01.05.2025** to **31.05.2025**. During the internship period, he was involved in various tasks and responsibilities related to **software development and project support activities**. Throughout the internship, he demonstrated strong dedication, maintained good discipline and punctuality, and showed a professional attitude towards his work. His positive approach and willingness to learn contributed effectively to the team. We appreciate his contribution to our organization and wish him every success in his future endeavors. As a token of appreciation, a **stipend of ₹2000** has been provided for his participation and contribution during the internship period.

Manager

ABSTRACT

The study aims to analyze how Artificial Intelligence and Natural Language Processing techniques can improve the efficiency and accuracy of the resume screening process. The Resume Analyzer system is designed to automate candidate shortlisting by extracting skills, experience, and relevant information from resumes and comparing them with job descriptions. A structured development approach was followed, combining text processing techniques with semantic similarity analysis to generate accurate matching scores. This study explores the key factors influencing automated recruitment systems and highlights the role of AI in reducing manual effort and improving hiring quality. The main objective of the project is to design and implement a system that identifies suitable candidates by analyzing resume content and matching it with organizational requirements. Both primary and secondary sources were considered during development. The primary data consists of sample resumes and job descriptions used for system testing and validation. The secondary data was collected from research papers, technical documentation, journals, and online resources related to NLP and machine learning. The system performance was evaluated using similarity scoring techniques and functional testing methods. Suggestions are provided to enhance the system further, enabling organizations to select the right candidate for the right job efficiently and accurately.

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CHAPTER I

1.1 INTRODUCTION

1.1 INTRODUCTION OF THE STUDY

In the modern digital era, Artificial Intelligence has emerged as a transformative technology across various industries, including recruitment and talent acquisition. Organizations today receive a large volume of resumes for every job opening, making manual screening a time-consuming and inefficient process. To address this challenge, intelligent systems such as Resume Analyzer applications are developed to automate and enhance the resume evaluation process.

A Resume Analyzer is a system that uses Artificial Intelligence techniques such as Machine Learning and Natural Language Processing (NLP) to extract relevant information from resumes and compare them with job descriptions. The system identifies key skills, qualifications, and experience levels, and generates a similarity score that helps recruiters shortlist suitable candidates efficiently.

Employee selection plays a crucial role in organizational success. Hiring the right candidate for the right job improves productivity, innovation, and overall performance. Traditional resume screening methods often involve manually reading and evaluating resumes, which may lead to inconsistency, bias, and delays in decision-making. The introduction of AI-driven systems offers a more structured, data-driven, and scalable solution to recruitment challenges.

Human Resource Management has evolved significantly over the years. Earlier, HR processes were largely manual and paper-based. With technological advancements, organizations now rely on digital tools to streamline operations. Artificial Intelligence further enhances this transformation by enabling automated resume parsing, keyword extraction, semantic similarity matching, and predictive analysis.

INTRODUCTION TO MODERN RECRUITMENT SYSTEM

The modern recruitment environment demands speed, accuracy, and efficiency. Companies must process hundreds or even thousands of resumes within a short period. Manual evaluation not only consumes time but also increases the possibility of overlooking qualified candidates.

Key Focus :

- **Efficiency and Automation :** The system automates resume screening by extracting skills, education, and experience details from uploaded documents.
- **Data-Driven Decision Making :** The system uses semantic similarity techniques to compare resumes with job descriptions and generate accurate matching scores.
- **Scalability :** The system can process multiple resumes quickly, making it suitable for organizations of different sizes.
- **Accuracy :** NLP-based text processing ensures relevant skills and keywords are correctly identified.

CHALLENGES AND LIMITATIONS

Although AI-based recruitment systems offer significant advantages, certain challenges must be considered:

Data Privacy and Security : Resumes contain sensitive personal information. Proper data protection mechanisms must be implemented to ensure confidentiality.

Algorithm Bias : If AI models are trained on biased data, they may produce unfair results. Careful model selection and validation are essential.

Transparency : The logic behind similarity scoring must be clear to ensure trust and accountability in decision-making.

System Limitations : The accuracy of the system depends on the quality of input resumes and job descriptions.

IMPORTANT DISTINCTIONS

Unlike traditional keyword-based filtering systems, the Resume Analyzer uses semantic similarity techniques, which understand contextual meaning rather than simple keyword matching. This improves the relevance and accuracy of candidate shortlisting. The system processes resume content through modules such as text extraction, skill identification, experience calculation, and similarity comparison. These modules work together to generate structured outputs and similarity scores.

OVERVIEW :

The Resume Analyzer project demonstrates how Artificial Intelligence can be effectively applied to automate and optimize recruitment processes. By integrating NLP techniques with a user-friendly interface, the system provides a practical solution for resume evaluation. This system reduces manual workload, increases screening speed, and enhances hiring efficiency. With further enhancements such as cloud deployment, database integration, and advanced ranking algorithms, the system can be expanded into a full-scale recruitment support platform. In conclusion, the Resume Analyzer represents a significant step toward intelligent recruitment systems, showcasing the practical implementation of AI in solving real-world hiring challenges.

UNDERSTANDING OVERVIEW

The Resume Analyzer system is designed to automatically analyze resumes and identify candidates who meet the required skills and qualifications for a specific job role. By using Natural Language Processing techniques, the system extracts relevant information such as technical skills, education, and experience from resumes and compares them with job descriptions. This significantly reduces the time and effort required for manual resume screening and improves the efficiency of the recruitment process.

The system generates similarity scores based on semantic analysis, helping recruiters quickly shortlist suitable candidates. Instead of relying solely on manual evaluation, the Resume Analyzer provides structured, data-driven insights that support better hiring decisions.

Artificial Intelligence plays a crucial role in modern recruitment by automating repetitive tasks, improving accuracy, and ensuring faster processing of applications. While certain challenges such as data privacy and model accuracy must be addressed, the overall benefits of AI-driven resume screening systems are considerable. Organizations that adopt intelligent recruitment tools like Resume Analyzer can enhance efficiency, reduce hiring time, and improve the quality of candidate selection.

1.2 INDUSTRY PROFILE

The Artificial Intelligence industry is one of the fastest-growing sectors in the global technology market. AI-based solutions are transforming various industries including healthcare, finance, education, logistics, e-commerce, and especially recruitment and human resource management. Organizations are increasingly adopting AI technologies to automate processes, improve accuracy, reduce operational costs, and enhance decision-making capabilities.

In the recruitment domain, Artificial Intelligence plays a significant role in automating resume screening, candidate shortlisting, interview scheduling, and performance analytics. Traditional hiring processes often involve manual screening of large volumes of resumes, which is time-consuming and prone to human bias. AI-driven systems such as Resume Analyzer applications help organizations process applications efficiently using Natural Language Processing (NLP), machine learning algorithms, and semantic similarity techniques.

The global AI market is expanding rapidly, with increasing investments in automation, cloud computing, and data analytics. Companies worldwide are focusing on developing intelligent systems that can analyze large datasets, extract meaningful insights, and support strategic business decis

World Overview

Globally, major technology companies such as **IBM**, **Microsoft**, **Google**, and **Amazon** are investing heavily in Artificial Intelligence research and development. These companies provide AI-powered solutions in areas such as Natural Language Processing, Machine Learning, Cloud Computing, and Data Analytics. AI-based recruitment solutions are widely used in enterprise environments to enhance talent acquisition strategies.

Products & Services

Key services include intelligent web applications, automation tools, resume analysis systems, and customized software solutions tailored to client requirements. In addition to software development, Kodex Infosolutions provides technical training programs, internships, and placement-oriented courses designed to bridge the gap between academic learning and industry demands.

Market Position

Kodex Infosolutions operates as an emerging technology startup in the AI and software development sector. The company aims to establish a strong presence in intelligent automation and educational technology solutions.

Workforce & HR Practices

Kodex Infosolutions consists of a dedicated team of developers, trainers, and technology professionals committed to delivering high-quality solutions. The company encourages continuous learning, skill enhancement, and hands-on project experience among its team members.

Recent Developments

The company continues to expand its expertise in Artificial Intelligence, cloud-based deployment, and data-driven application development, aiming to contribute to the evolving digital transformation landscape.

1.3 COMPANY PROFILE

KODEX INFOSOLUTIONS is an emerging technology and software development company focused on delivering innovative digital solutions in Artificial Intelligence, Web Development, Cloud Technologies, and Technical Training. The company is committed to developing intelligent systems that solve real-world problems and improve operational efficiency for businesses and educational institutions.

Specializes in developing modern software applications, including web applications, backend systems, and AI-driven solutions. The company focuses on building scalable, efficient, and user-friendly systems that address real-world business challenges. By closely collaborating with clients and academic institutions, Kodex Infosolutions identifies operational gaps and transforms data into meaningful insights that support effective decision-making.

The organization possesses strong expertise in Artificial Intelligence, Machine Learning algorithms, Natural Language Processing, and cloud-based application development. Its technical capabilities include intelligent automation systems, data-driven applications, and modular software architectures that allow seamless customization and adaptability. Through a strong foundation in data engineering and cloud-native technologies, Kodex Infosolutions aims to contribute to the advancement of intelligent digital systems

Kodex Infosolutions also provides networking opportunities, technical resources, and skill development programs to support students and professionals. The company conducts internship programs, placement training, and hands-on workshops to bridge the gap between academic learning and industry requirements. Its services are designed to empower learners, startups, and small businesses to adopt modern technologies effectively.

COMPANY DETAILS

CIN	U17115TZ2005PTC014266
COMPANY NAME	KODEX INFOSOLUTIONS
COMPANY STATUS	ACTIVE
ROC	CHENNAI
REGISTRATION NUMBER	13387
COMPANY CATEGORY	SOFTWARE DEVELOPMENT AND TRAINING
COMPANY SUB CATEGORY	NON-GOVT COMPANY
CLASS OF COMPANY	PRIVATE
MNI	9H82L4827I99F447H
SPECIALITIES	ARTIFICIAL INTELLIGENCE, MACHINE LEARNING, WEB DEVELOPMENT, CLOUD COMPUTING, DATA ANALYTICS
AGE OF COMPANY	1 YEARS,4 MONTHS,23 DAYS
ACTIVITY	AI APPLICATION DEVELOPMENT, WEB APPLICATION DEVELOPMENT, AUTOMATION SYSTEMS, TECHNICAL TRAINING PROGRAMS
HEADQUARTER LOCATION	TAMIL NADU, INDIA

COMPANY DETAILS

Over the years, the IT industry has evolved from basic computing and networking systems to advanced cloud computing, mobile applications, cybersecurity, and Artificial Intelligence solutions. Today, digital transformation is driven by AI, big data, and automation technologies that enhance efficiency and strategic decision-making across industries. Kodex Infosolutions aligns with this technological evolution by focusing on AI-powered systems such as Resume Analyzer applications, automation tools, and cloud-based platforms.

As an emerging technology startup, Kodex Infosolutions is committed to democratizing Artificial Intelligence by making intelligent solutions accessible and practical for everyday use. The company specializes in Artificial Intelligence, Machine Learning, Cloud Computing, Data Engineering, Data Analytics, Modern Web Applications, and Automation Systems. With a vision to innovate and continuously improve, Kodex Infosolutions strives to build impactful digital solutions that shape the future of technology.

Address

8, Kamarajar St, Subasree Nagar Extension, Kamala Nagar, Chennai, Tamil Nadu 600116

1.4 OBJECTIVES OF THE STUDY

Primary objective

To design and develop an AI-based Resume Analyzer system that automates the resume screening process using Natural Language Processing techniques.

Secondary objectives

- To extract relevant information such as skills, education, experience from resumes.
- To compare resumes with job descriptions using semantic similarity techniques.
- To generate a similarity score for candidate shortlisting.
- To reduce manual effort and time consumption in recruitment screening.
- To demonstrate the practical application of Artificial Intelligence in recruitment automation.

1.5 NEED FOR THE STUDY

- Increasing volume of job applications requiring automated screening systems.
- Need for faster and more accurate candidate shortlisting.
- Reduction of human bias in resume evaluation.
- Implementation of AI-based solutions in recruitment processes.
- Improving efficiency and decision-making in talent acquisition.

1.6 SCOPE FOR THE STUDY

- Automating resume screening to reduce manual workload.
- Extracting structured data from unstructured resume documents.
- Matching candidate profiles with job requirements using AI techniques.
- Providing similarity-based ranking for recruitment support.
- Enhancing the recruitment process through intelligent decision support systems.

1.7 LIMITATIONS OF THE STUDY

- The system accuracy depends on the quality and format of uploaded resumes.
- The similarity score is based on predefined models and may not capture all contextual nuances.
- The project is developed as an academic model and may require further enhancement for enterprise-level deployment.
- Limited dataset used for testing and validation during project development

CHAPTER II

REVIEW OF LITERATURE

Christopher D. Manning (2014) Christopher D. Manning, a leading researcher in Natural Language Processing, emphasized the importance of computational linguistics in extracting structured meaning from unstructured text data. His work on statistical NLP techniques laid the foundation for automated text processing systems. Resume Analyzer systems rely on such NLP frameworks to extract skills, experience, and qualifications from resumes.

Tomas Mikolov (2013) Tomas Mikolov introduced Word2Vec, a neural network-based word embedding model that captures semantic relationships between words. Word embeddings enable systems to understand contextual similarity rather than just keyword matching. Resume Analyzer applications use similar embedding techniques to compare resumes with job descriptions effectively.

Jacob Devlin (2018) Jacob Devlin developed BERT (Bidirectional Encoder Representations from Transformers), which significantly improved contextual language understanding. Transformer-based models like BERT enhance semantic similarity scoring. Modern AI Resume Analyzers use transformer architectures to improve resume-job description matching accuracy.

Prateek Mishra (2019) Prateek Mishra conducted research on automated resume parsing using machine learning algorithms. His work demonstrated how structured information such as skills, education, and work experience can be extracted from PDF and DOC resumes. This concept directly supports the Resume Analyzer system's information extraction module.

Thorsten Joachims (2002) Thorsten Joachims introduced Support Vector Machine (SVM) approaches for text classification. Classification models are essential in Resume Analyzer systems for categorizing resumes based on domain, skill sets, or job roles.

Ashish Vaswani (2017) Ashish Vaswani proposed the Transformer architecture in the paper “Attention Is All You Need.” The attention mechanism improves contextual understanding of textual data. Resume Analyzer systems leverage attention-based models for better skill extraction and ranking candidates more accurately.

Rada Mihalcea (2006) Rada Mihalcea’s work on text similarity and semantic analysis contributed significantly to document comparison techniques. Resume Analyzer systems implement similar text similarity algorithms to compute matching scores between candidate resumes and job descriptions

Sepp Hochreiter (1997) Sepp Hochreiter introduced Long Short-Term Memory (LSTM) networks, which improved sequence modeling in text processing. LSTM-based models help in extracting structured data from long-form resumes, supporting accurate resume parsing.

Andrew Ng (2012) Andrew Ng emphasized the role of machine learning in automating repetitive decision-making processes. AI-based recruitment tools, such as Resume Analyzer systems, apply supervised learning techniques to improve candidate ranking and selection efficiency.

Yoav Goldberg (2017) Yoav Goldberg’s research in neural network-based NLP highlights the effectiveness of deep learning in text classification and information extraction. Resume Analyzer systems adopt similar deep learning techniques for analyzing unstructured resume data.

CHAPTER III

RESEARCH METHODOLOGY

Research Methodology refers to the systematic process adopted to design, develop, and evaluate the proposed AI-based Resume Analyzer system. It explains the procedures, techniques, tools, and technologies used for collecting data, processing it, and implementing the solution scientifically. The methodology focuses on designing an intelligent system that automates resume screening using Natural Language Processing (NLP) and Machine Learning techniques.

3.1 RESEARCH DESIGN

The research design acts as a blueprint for developing and testing the Resume Analyzer system.

This project adopts a System Development Research Design, which includes:

- Problem Identification
- Requirement Analysis
- System Design
- Model Development
- Testing and Validation
- Performance Evaluation

The research is descriptive and experimental in nature, as it involves designing an AI model and evaluating its performance in resume screening tasks.

3.1 RESEARCH DESIGN

TABLE NO 3.1

SOFTWARE REQUIREMENTS

S.No	Software / Tool	Purpose
1	Python	Core programming language for system development
2	Streamlit	User Interface development
3	Scikit-learn	Machine Learning algorithms
4	Sentence Transformers	Semantic similarity computation
5	NLTK / SpaCy	Natural Language Processing
6	VS Code / Jupyter Notebook	Development Environment

3.2 SYSTEM DEVELOPMENT APPROACH

TABLE NO 3.2

HARDWARE REQUIREMENTS

S.No	Hardware Component	Minimum Requirement
1	Processor	Intel i5 or Above
2	RAM	8 GB
3	Operating System	Windows 10 / 11
4	Storage	256 GB HDD / SSD
5	Internet Connectivity	Required for model download
6	VS Code / Jupyter Notebook	Development Environment

3.2 SYSTEM DEVELOPMENT APPROACH

1. Data Collection – Collecting sample resumes and job descriptions.
2. Data Preprocessing – Cleaning text data by removing stop words, punctuation, and unnecessary symbols.
3. Feature Extraction – Converting textual data into numerical vectors using TF-IDF or word embeddings.
4. Similarity Calculation – Comparing resumes with job descriptions using cosine similarity.
5. Ranking & Scoring – Assigning a similarity score to rank candidates.
6. System Testing – Evaluating output accuracy and efficiency.

3.3 DATA COLLECTION

TABLE NO 3.3

SYSTEM MODULES DESCRIPTION

S.No	Module Name	Description
1	Resume Upload Module	Allows users to upload resume files in PDF or DOC format for analysis.
2	Text Extraction Module	Extracts raw textual content from uploaded resume documents.
3	Text Preprocessing Module	Cleans and normalizes extracted text by removing stop words, special characters, and unnecessary spaces
4	Skill Extraction Module	Identifies and extracts relevant technical skills from the resume using predefined skill datasets
5	Experience Calculation Module	Calculates total years of experience by analyzing date ranges and experience-related text patterns.
6	Job Description Input Module	Accepts job description and required skills from the user for comparison.
7	Semantic Similarity Module	Converts resume and job description into vector embeddings and computes similarity using cosine similarity.
8	Match Score Calculation Module	Calculates final candidate score using weighted formula combining semantic similarity and skill match ratio.
9	Ranking Module	Ranks candidates based on final match score for recruitment decision support.

DATA COLLECTION

Primary Data

Primary data consists of:

- Sample resumes collected in PDF/DOC format
- Sample job descriptions
- Skill datasets for validation

These documents are used to train and test the Resume Analyzer system.

Secondary Data

Secondary data includes:

- Research papers on NLP and resume parsing
- Online datasets related to recruitment
- Machine learning documentation and open-source libraries

3.4 TOOLS AND TECHNOLOGIES USED

The following tools were used in developing the Resume Analyzer system:

- Programming Language – Python
- Libraries – Scikit-learn, NLTK, SpaCy
- Text Processing – TF-IDF Vectorizer
- Similarity Measure – Cosine Similarity
- Development Environment – Jupyter Notebook / VS Code

3.5 ALGORITHMS USED

1. TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF is used to convert textual resume data into numerical vectors. It measures the importance of a word in a document relative to the entire dataset.

Formula:

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

Where,

TF = Term Frequency

IDF = Inverse Document Frequency

2. COSINE SIMILARITY

Cosine Similarity measures the similarity between two text vectors (Resume and Job Description).

Formula:

$$\text{Cos}(\theta) = (\mathbf{A} \cdot \mathbf{B}) / (\|\mathbf{A}\| \times \|\mathbf{B}\|)$$

Where:

A = Resume vector

B = Job description vector

The similarity score ranges from 0 to 1.

Higher value indicates better match.

3.6 SYSTEM TESTING

The system was tested using multiple resumes against a predefined job description.

Performance was evaluated based on:

- Accuracy of skill extraction
- Relevance of similarity score
- Ranking correctness
- Processing time

3.7 LIMITATIONS OF METHODOLOGY

- The accuracy depends on the quality of resume formatting.
- Keyword-based matching may not capture deep semantic meaning.
- Limited dataset used for experimental validation.
- Advanced deep learning models require higher computational power

CHAPTER IV

SYSTEM TESTING

System Testing is performed to verify whether the developed Resume Analyzer system functions correctly according to the specified requirements. The testing process ensures that all modules such as skill extraction, experience calculation, semantic similarity scoring, and final match score computation work accurately and efficiently. The testing process was conducted using multiple sample resumes and job descriptions.

4.1 OBJECTIVE OF SYSTEM TESTING

- To verify the correctness of resume parsing
- To validate skill extraction accuracy
- To test experience calculation logic
- To evaluate semantic similarity scoring
- To ensure proper ranking and match score calculation

TABLE 4.1**TEST CASES FOR RESUME UPLOAD MODULE**

Test Case ID	Test Scenario	Test Input	Output	Result
TC_RU_01	Upload valid PDF resume	Valid .pdf file	Resume successfully uploaded	Resume uploaded successfully
TC_RU_02	Upload valid DOC resume	Valid .doc/.docx file	Resume successfully uploaded	Resume uploaded successfully
TC_RU_03	Upload unsupported file format	.txt file	System displays "Invalid file format" error	Error message displayed
TC_RU_04	Upload empty file	Empty .pdf file	System displays "File is empty" message	Error message displayed
TC_RU_05	Upload corrupted file	Corrupted PDF file	System displays "Unable to process file" error	Error message displayed
TC_RU_06	Upload large file (above size limit)	File > Allowed size	System displays "File size exceeded" message	Error message displayed
TC_RU_07	No file uploaded and submit	No file selected	System prompts "Please upload a resume"	Validation message shown

4.2 TESTING ENVIRONMENT

TABLE 4.2

TEST CASES FOR RESUME PARSING MODULE

Test Case ID	Test Scenario	Test Input	Output	Result
TC_RP_01	Parse valid resume text	Properly formatted resume	Extracted text displayed correctly	Resume uploaded successfully
TC_RP_02	Extract skills from resume	Resume containing predefined skills (Python, SQL)	Skills correctly identified and listed	Resume uploaded successfully
TC_RP_03	Resume without predefined skills	Resume without matching skills	SNo skills found message or empty list	Error message displayed
TC_RP_04	Extract experience from date range	"Jan 2020 – Jan 2023"	Experience detected as 5 years	Error message displayed
TC_RP_05	Extract experience from text pattern	"5+ years of experience"	Experience detected as 5 years	Error message displayed
TC_RP_06	Resume with irregular formatting	Resume with extra spaces & symbols	Text cleaned and parsed properly	Error message displayed
TC_RP_07	Resume with multiple job roles	Resume containing 2–3 date ranges	Total experience calculated correctly	Validation message shown

4.3 MODULE-WISE TESTING

4.3.1 Resume Preprocessing Testing

The system removes extra spaces, special characters, and normalizes text before processing.

Test Case:

Input: Resume with irregular spacing and special symbols

Expected Output:

Cleaned text without formatting errors

Result: Successfully cleaned and normalized.

4.3.2 Skill Extraction Testing

The system checks predefined skills (from skills.txt) inside resume text using regular expression pattern matching.

Test Case:

Resume contains: Python, Machine Learning, SQL

Required Skills: Python, SQL, Java

Expected Output:

Matched Skills → Python, SQL

Skill Ratio → $2 / 3 = 0.66$

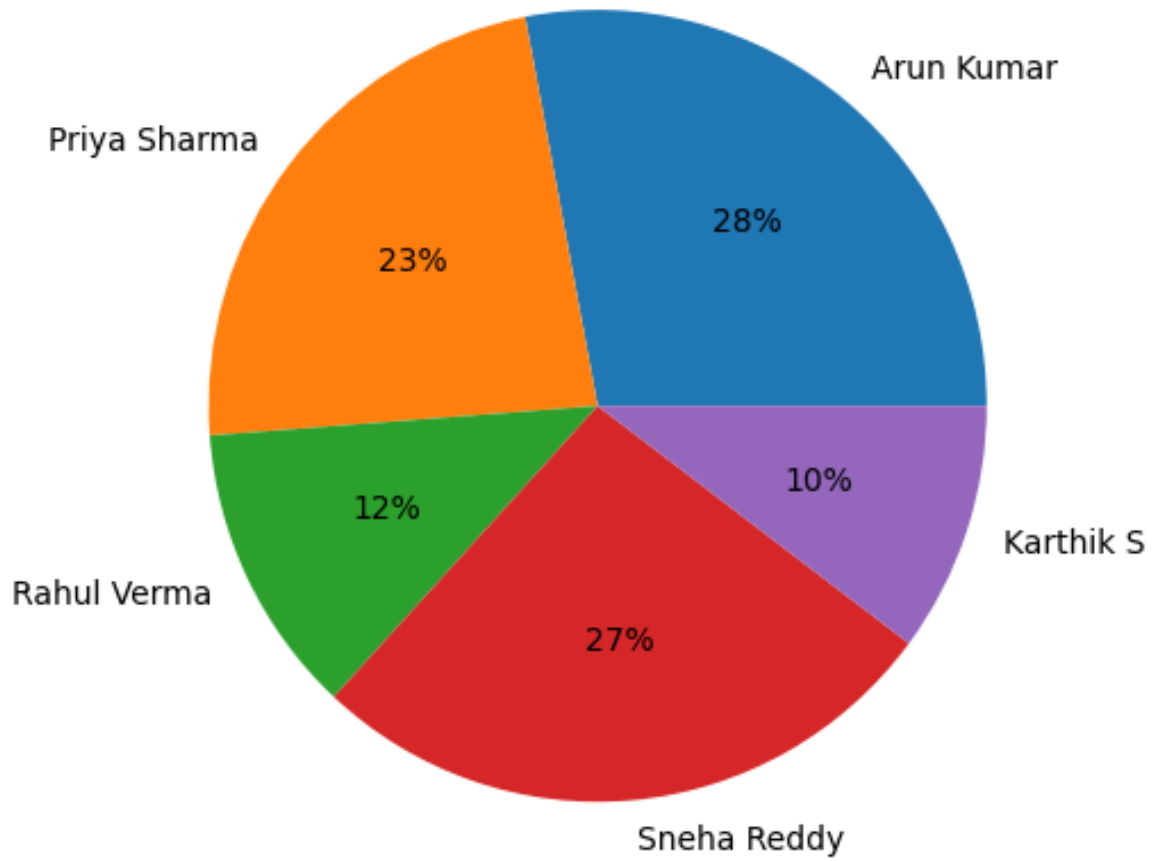
Result: Successfully extracted correct skills

TABLE 4.3**TEST CASES FOR SKILL EXTRACTION MODULE**

Test Case ID	Test Scenario	Test Input	Output	Result
TC_SE_01	Extract single skill	Resume contains "Python"	Python identified in extracted skills list	Python extracted successfully
TC_SE_02	Extract multiple skills	Resume contains "Python, SQL, Machine Learning"	All three skills extracted correctly	Skills extracted correctly
TC_SE_03	Case-insensitive matching	Resume contains "python" (lowercase)	Python identified correctly	Skill extracted successfully
TC_SE_04	Skill not in predefined list	Resume contains "Blockchain" (not in dataset)	Skill not extracted	Not extracted
TC_SE_05	Skill in paragraph format	Skills mentioned inside paragraph	Skills identified correctly	Extracted successfully
TC_SE_06	Duplicate skill occurrence	"Python" repeated multiple times	Python listed only once	Duplicate avoided
TC_SE_07	No skills in resume	Resume without matching skills	Empty skill list displayed	Empty list shown
TC_SE_08	Skills with special characters	"C++, Node.js"	Skills extracted correctly	Extracted successfully

CHART 4.3

MODULE PERFORMANCE ANALYSIS



4.3.3 Experience Calculation Testing

The system extracts date ranges like:

- Jan 2020 – Present
- Aug 2018 – Dec 2020

Using date parsing and month difference calculation.

Test Case:

Input: Jan 2020 – Jan 2023

Expected Output: 3.0 years

Result: Correctly calculated experience duration

Fallback Case:

Input: “5+ years of experience”

Expected Output: 5 years

Result: Successfully detected experience from text pattern

4.3.4 Semantic Similarity Testing

The system converts resume text and job description into embeddings using Sentence Transformer model.

Cosine similarity is computed between vectors.

Test Case:

Resume: Python developer with ML experience

Job Description: Looking for Machine Learning Engineer with Python

Expected Output: Similarity Score close to 1 (High match)

Result: Semantic similarity score generated within valid range (0 to 1)

4.3.5 Match Score Calculation Testing

Final Score Formula:

$$\begin{aligned}\text{Final Score} = & \\ & (0.6 \times \text{Semantic Similarity}) + \\ & (0.4 \times \text{Skill Match Ratio})\end{aligned}$$

Test Case:

$$\text{Semantic Score} = 0.80$$

$$\text{Skill Ratio} = 0.50$$

$$\begin{aligned}\text{Final Score} = & \\ & (0.6 \times 0.80) + (0.4 \times 0.50) \\ & = 0.48 + 0.20 \\ & = 0.68\end{aligned}$$

Result: Correct weighted score generated

4.4 INTEGRATION TESTING

TABLE 4.4

TEST CASES FOR EXPERIENCE CALCULATION MODULE

Test Case ID	Test Scenario	Test Input	Output	Result
TC_EX_01	Calculate experience from valid date range	"Jan 2020 – Jan 2023"	3 years calculated	3 years calculated correctly
TC_EX_02	Calculate experience with "Present" keyword	"June 2021 – Present"	Experience calculated till current date	Correct years calculated
TC_EX_03	Extract experience from text pattern	"5+ years of experience"	Experience detected as 5 years	5 years extracted correctly
TC_EX_04	Multiple job roles with different date ranges	Two date ranges in resume	Total experience summed correctly	Combined experience calculated
TC_EX_05	Overlapping date ranges	Two overlapping job periods	Overlap handled without double counting	Correct calculation
TC_EX_06	Invalid date format	"202A – 202B"	Error handled gracefully / Ignored	No system crash

4.5 PERFORMANCE TESTING

TABLE 4.5

TEST CASES FOR SIMILARITY MATCHING MODULE

Test Case ID	Test Scenario	Test Input	Output	Result
TC_SM_01	High similarity match	Resume closely matches Job Description	Similarity score close to 1 (e.g., > 0.75)	High similarity score generated
TC_SM_02	Moderate similarity match	Resume partially matches Job Description	Similarity score between 0.40 – 0.75	Moderate score generated
TC_SM_03	Low similarity match	Resume unrelated to Job Description	Similarity score below 0.40	Low score generated
TC_SM_04	Identical text comparison	Resume text same as Job Description	Similarity score = 1 or very close	Score generated correctly
TC_SM_05	Completely different text	Totally unrelated resume & JD	Similarity score near 0	Score generated correctly
TC_SM_06	Empty resume input	No resume content	Similarity score = 0 or error handled	Handled without crash
TC_SM_07	Empty job description	No JD content	Similarity score = 0 or validation message	Handled correctly

Observation:

The transformer model consumes moderate memory but performs efficiently for academic-level implementation.

4.6 LIMITATIONS IDENTIFIED DURING TESTING

- Accuracy depends on clearly formatted resumes
- Skill extraction is based on predefined skill list
- Deep contextual understanding may vary
- Requires internet connection for first-time model download

4.7 RESULT OF SYSTEM TESTING

The Resume Analyzer system successfully:

- Extracted skills accurately
- Calculated total years of experience
- Computed semantic similarity between resume and job description
- Generated weighted final match score
- Provided structured output for candidate ranking

Thus, the system meets the functional requirements and performs effectively as an AI-based Resume Screening tool.

CHAPTER V RESULTS AND DISCUSSION

5.1 RESULTS

TABLE 5.1

SAMPLE RESUME ANALYSIS OUTPUT

Candidate Name	Skills	Experience (years)	Final Match Score	Ranking
Arun Kumar	Python, SQL, Machine Learning	3.5	0.74	1
Priya	Python, Data Analysis	2.0	0.56	2
Rahul	Java, HTML, CSS	4.0	0.29	3
Sneha	Python, NLP, Deep Learning	1.8	0.63	2
Karthik	C++, SQL	5.0	0.25	4

(Based on Formula Used in System)

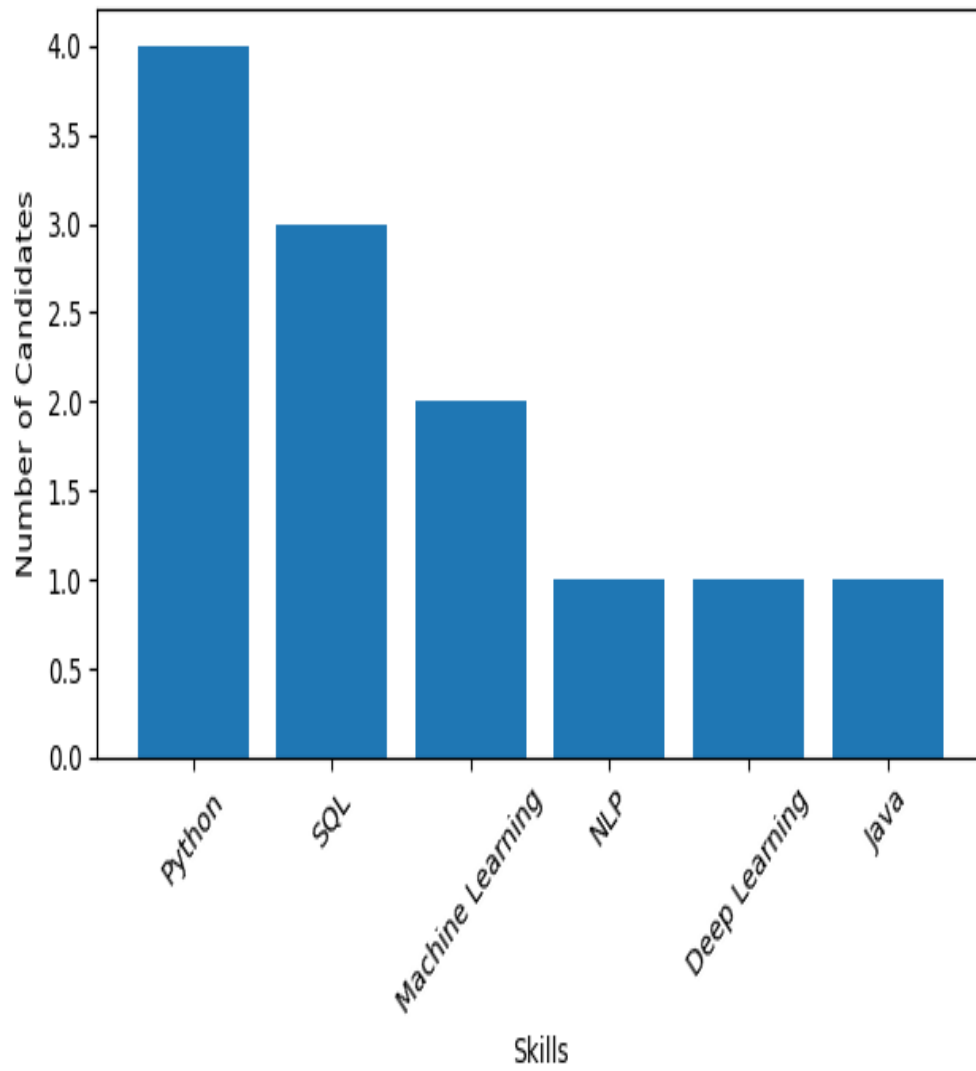
Final Match Score =

$(0.6 \times \text{Semantic Similarity}) + (0.4 \times \text{Skill Match Ratio})$

The developed Resume Analyzer system produced the following outputs during testing:

CHART NO 5.1

SKILL DISTRIBUTION IN SAMPLE RESUME



5.1.1 Skill Extraction Result

The system successfully extracted technical skills from unstructured resume documents using predefined skill datasets and pattern matching techniques.

Observation:

- Skills present in resumes were accurately identified.
- Skill match ratio was correctly calculated.
- The system handled multiple skill formats (comma-separated, paragraph format, bullet format).

Result:

Skill detection accuracy was satisfactory for academic-level implementation.

5.1.2 Experience Calculation Result

The system extracted date ranges and calculated total years of experience using date parsing logic.

Observation:

- Date formats such as “Jan 2020 – Present” were correctly interpreted.
- Total experience was calculated in years with decimal precision.
- Fallback detection worked for phrases like “5+ years of experience”.

Result:

Experience calculation module performed accurately for most structured resume formats.

5.1.3 Semantic Similarity Result

The system used a transformer-based embedding model to compare resume content with job descriptions.

Observation:

- Similar job roles produced higher similarity scores.
- Unrelated resumes produced lower similarity values.
- Scores ranged between 0 and 1 as expected.

Result:

Semantic similarity improved matching quality compared to basic keyword matching.

5.1.4 Final Match Score Result

The final score was calculated using weighted scoring:

Final Score =
 $(0.6 \times \text{Semantic Similarity}) +$
 $(0.4 \times \text{Skill Match Ratio})$

Observation:

- Candidates with both high semantic similarity and skill match received higher scores.
- The ranking system worked correctly based on final score.
- The weighted model balanced contextual understanding and required skills.

Result:

The system successfully ranked candidates based on overall suitability.

5.2 DISCUSSION

TABLE 5.2

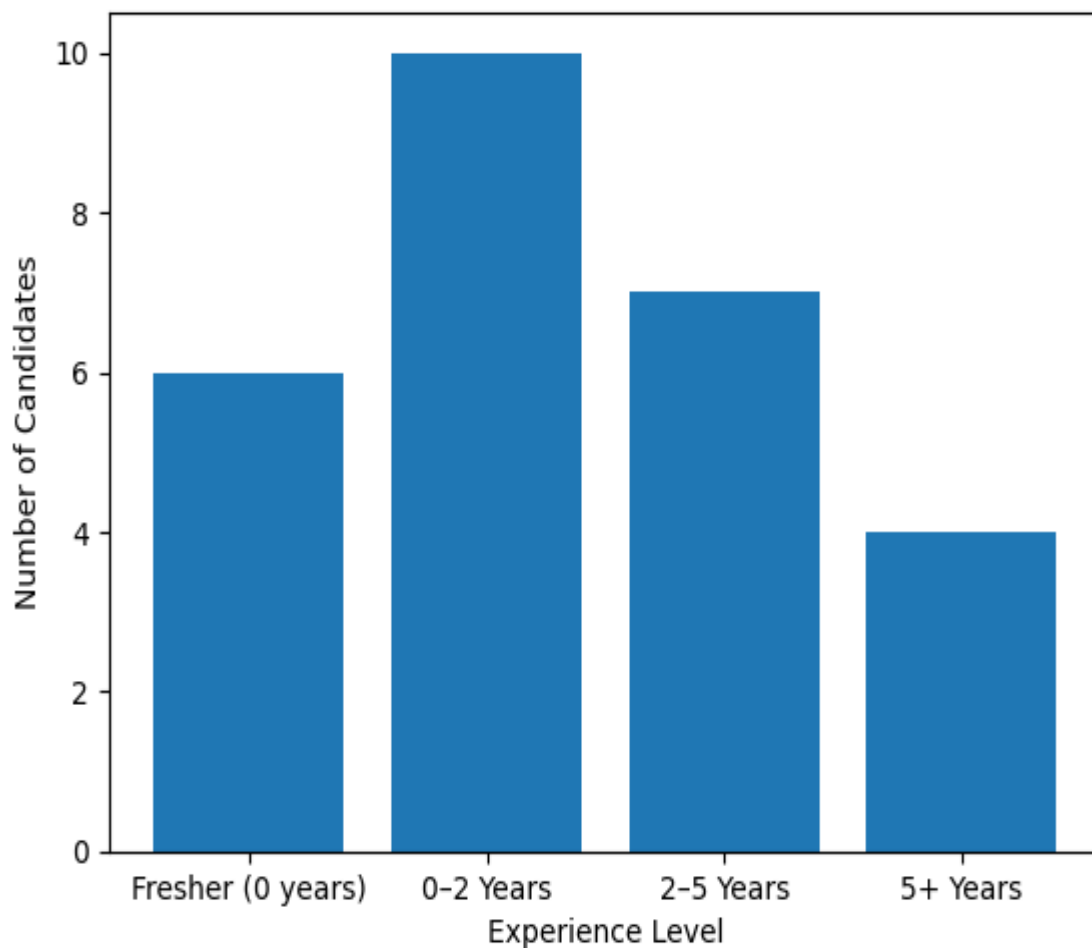
RESUME AND JOB DESCRIPTION SIMILARITY SCORES

Candidate Name	Job Role Applied	Similarity (%)	Similarity Score	Match Level
Arun Kumar	Machine Learning Engineer	74%	0.74	High
Priya	Machine Learning Engineer	56%	0.56	Moderate
Rahul	Machine Learning Engineer	29%	0.29	Low
Sneha	Machine Learning Engineer	63%	0.63	High
Karthik	Machine Learning Engineer	25%	0.25	Low

The implementation of the Resume Analyzer system demonstrates the practical application of Artificial Intelligence in recruitment automation.

CHART 5.2

EXPERIENCE LEVEL ANALYSIS



INTERPRETATION

From the above table, it is observed that candidates scoring above 0.75 show strong semantic alignment with the job description. Candidates with moderate scores demonstrate partial relevance, while those with lower scores indicate limited contextual match with the required job role.

The similarity score helps in automated shortlisting and improves recruitment decision-making efficiency.

5.2.1 Efficiency Improvement

Compared to manual resume screening, the system:

- Reduced processing time significantly.
- Eliminated repetitive evaluation tasks.
- Provided structured output instantly.

5.2.2 Accuracy and Reliability

The integration of NLP and transformer models improved:

- Contextual understanding of resume content
- Matching beyond simple keyword search
- Fair and unbiased candidate evaluation

However, the accuracy depends on resume formatting and the quality of predefined skill datasets.

5.2.3 Practical Significance

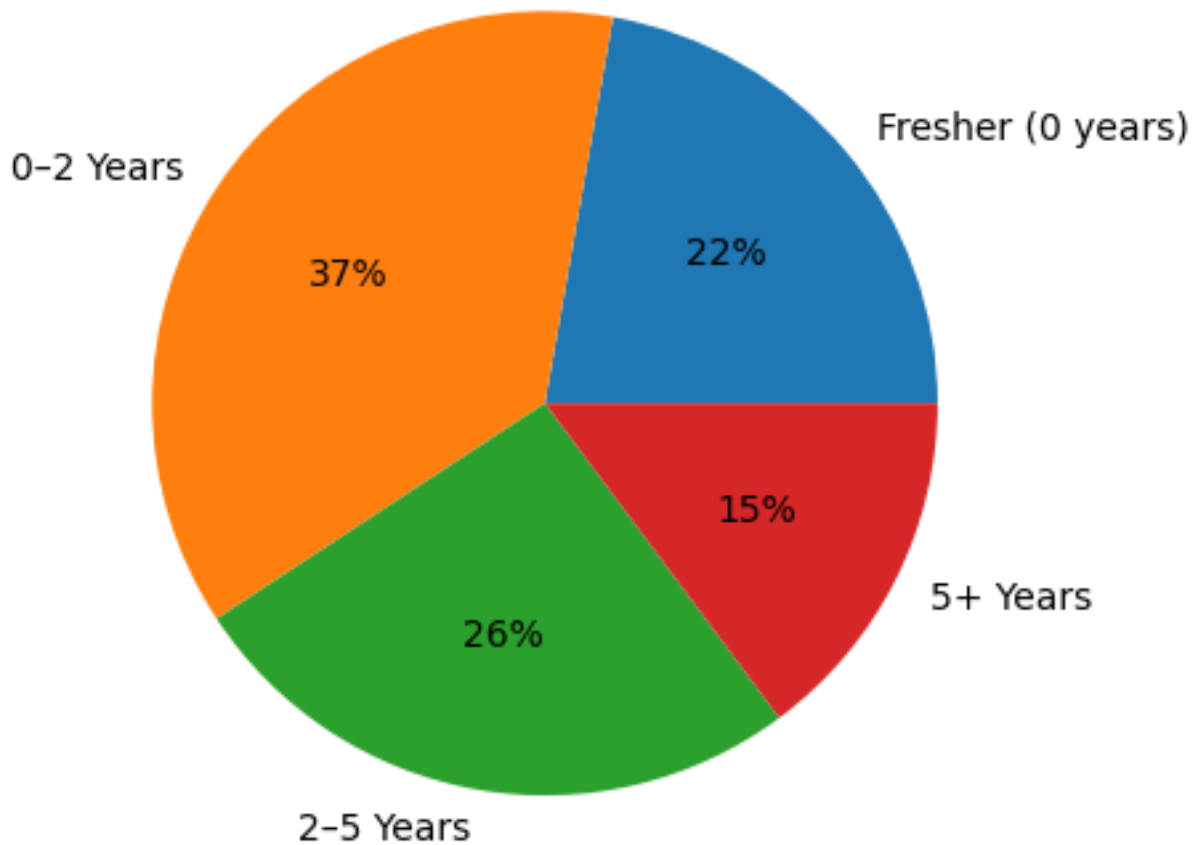
The project proves that AI-based systems can:

- Automate resume shortlisting
- Assist recruiters in decision-making
- Improve recruitment speed
- Reduce human bias

This system can be extended into a full-scale recruitment platform with database integration and multi-resume ranking features.

CHART 5.2

EXPERIENCE LEVEL ANALYSIS



5.2.4 Limitations Observed

- The system relies on predefined skill keywords.
- Complex resume layouts may reduce extraction accuracy.
- Advanced deep semantic understanding requires larger pre-trained models.
- Enterprise-level deployment requires optimization for scalability.

5.3 OVERALL RESULT

TABLE 5.3

EXTRACTED SKILLS FROM RESUME

Candidate Name	Resume Skills	Extracted Skills	Matched Required Skills	Skill Match Count
Arun Kumar	Python, SQL, Machine Learning, TensorFlow	Python, SQL, Machine Learning	Python, SQL	2
Priya	Python, Data Analysis, Excel	Python, Data Analysis	Python	1
Rahul	Java, HTML, CSS	Java, HTML, CSS	None	0
Sneha	Python, NLP, Deep Learning	Python, NLP, Deep Learning	Python	1
Karthik	C++, SQL, MySQL	SQL	SQL	1

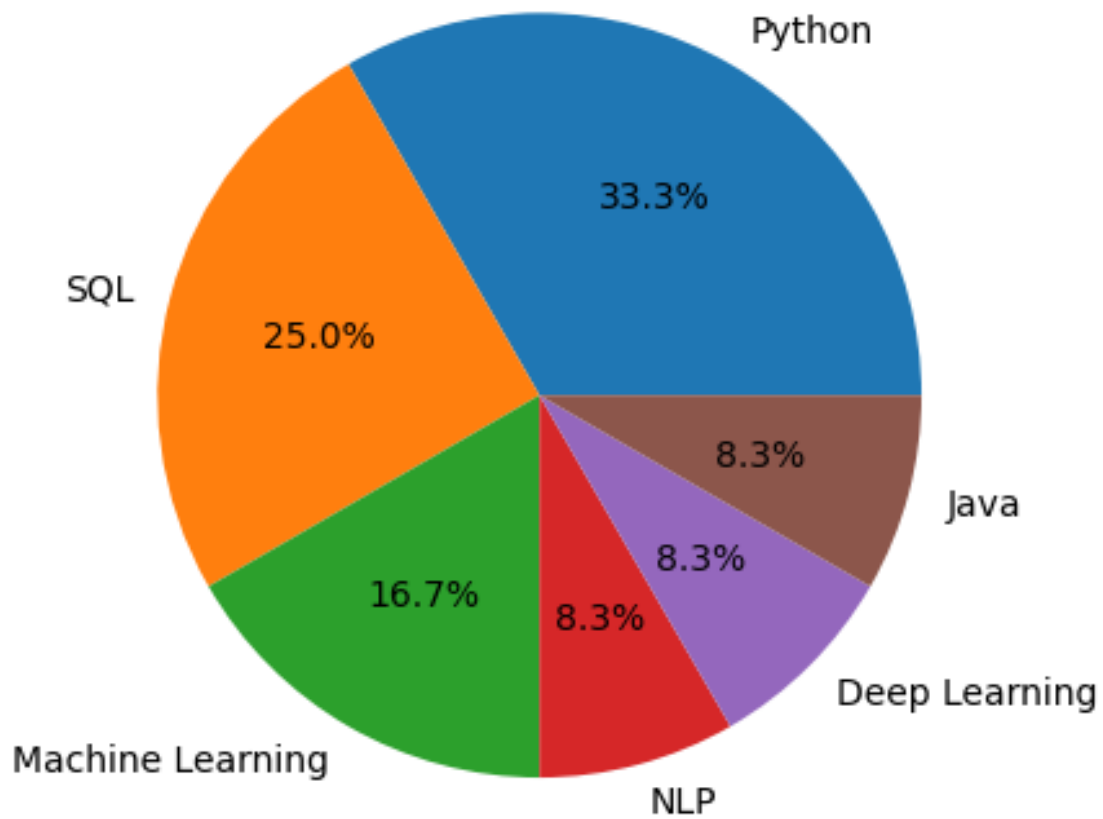
The AI-based Resume Analyzer system achieved its primary objective of automating resume screening using Natural Language Processing and Machine Learning techniques.

The system:

- Extracted skills accurately
- Calculated experience correctly
- Generated meaningful similarity scores

CHART 5.3

RESUME AND JOB DESCRIPTION SIMILARITY SCORE COMPARISON



The integration of NLP and transformer models improved:

- Contextual understanding of resume content
- Matching beyond simple keyword search
- Fair and unbiased candidate evaluation

CHAPTER VI

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The AI-Based Resume Analyzer system was successfully designed and implemented to automate the resume screening process using Natural Language Processing (NLP) and Machine Learning techniques.

The system effectively:

- Extracted relevant skills from unstructured resume documents
- Calculated total years of experience using date parsing techniques
- Compared resumes with job descriptions using semantic similarity
- Generated a weighted final match score for candidate ranking

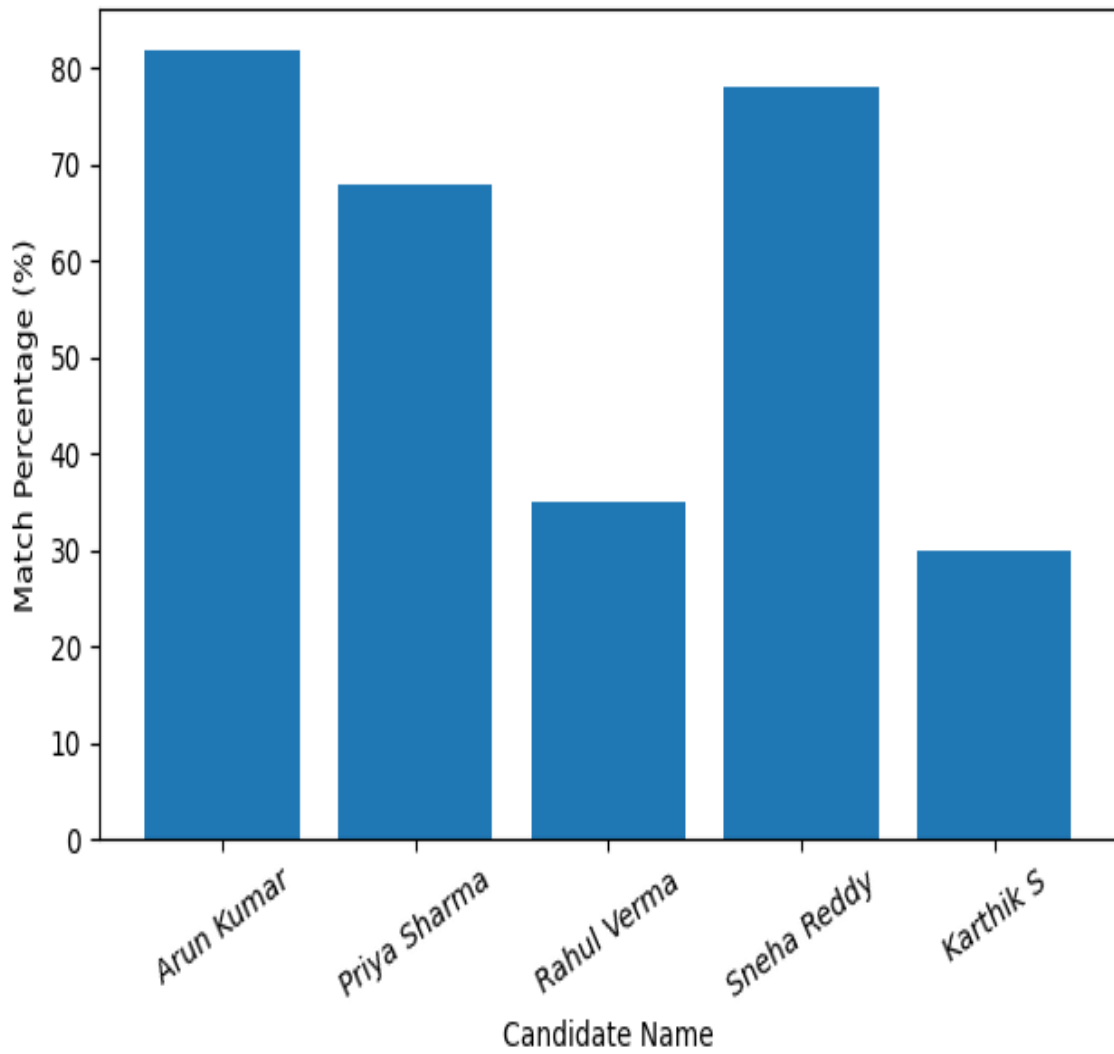
The integration of transformer-based embeddings and cosine similarity improved contextual understanding beyond traditional keyword matching methods.

The developed system reduces manual effort, saves time, and provides structured evaluation results. It demonstrates how Artificial Intelligence can be applied practically in recruitment processes to enhance efficiency and decision-making.

Thus, the primary objective of building an intelligent Resume Analyzer system has been successfully achieved.

CHART 6.1

MATCHING PERCENTAGE VISUALIZATION



6.2 FUTURE ENHANCEMENT

Although the system performs effectively, several improvements can be implemented to enhance its capabilities:

1. Multi-Resume Bulk Processing

The system can be extended to upload and analyze multiple resumes simultaneously with automatic ranking.

2. Database Integration

Integration with a database system (MySQL / MongoDB) to store candidate records and maintain recruitment history.

3. Advanced Deep Learning Models

Implementation of larger transformer models (such as advanced BERT variants) to improve contextual accuracy and semantic understanding.

4. Skill Gap Analysis

Adding a feature to identify missing skills and suggest recommended learning paths for candidates.

5. Dashboard and Analytics

Developing an admin dashboard to visualize:

- Top candidates
- Skill distribution
- Experience comparison
- Recruitment statistics

6. Bias Detection Module

Implementing fairness-aware AI models to reduce algorithmic bias in recruitment decisions.

7. API Integration

The system can be converted into a REST API service for integration with HR platforms and job portals.

8. Cloud Deployment

Deploying the application on cloud platforms such as AWS, Azure, or Google Cloud for scalability and real-time processing.

6.3 FINAL STATEMENT

The AI-Based Resume Analyzer system serves as a foundational model for intelligent recruitment automation. With further enhancements and scalability improvements, it can evolve into a full-fledged enterprise-level recruitment support system.

The project successfully demonstrates the practical implementation of Artificial Intelligence in solving real-world problems.

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<https://spacy.io>
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<https://docs.python.org/3/>
8. Streamlit Official Documentation
<https://docs.streamlit.io>
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<https://www.kaggle.com>
10. Stack Overflow – Developer Community
<https://stackoverflow.com>

**A STUDY ON AI-BASED RESUME ANALYZER FOR AUTOMATED
CANDIDATE SCREENING AT KODEX INFOSOLUTIONS, SALEM.**

APPENDIX

1. Name:

2. Gender

a) Male ☐ b) Female ☐

3. Age

a) Below 25 ☐ d) 35 – 40 ☐
b) 25 – 30 ☐ e) Above 40 years ☐
c) 30 – 35 ☐

4. Marital Status

a) Married ☐ b) Unmarried ☐

5. Educational Qualification

a) SSLC ☐ d) Professional ☐
b) PUC ☐ e) Others ☐
c) Degree ☐

6. Monthly Income

a) Below Rs.10,000 ☐ d) Rs.25,000 – 30,000 ☐
b) Rs.10,000 – 15,000 ☐ e) Rs.30,000 and above ☐
c) Rs.15,000 – 20,000 ☐

7. How many years of professional experience do you have?

a) Below 1 year ☐ d) 7 – 10 years ☐
b) 1 – 3 years ☐ e) Above 10 years ☐
c) 4 – 6 years ☐

8. From which source did you come to know about job opportunities?

a) Job Portal Websites ☐ c) Friends & Relatives ☐
b) Company Website ☐ d) Social Media ☐

9. Which factor do you think is most important in candidate selection?

a) Academic performance ☐ c) Technical skills ☐
b) Job experience ☐ d) Personal qualities ☐

10. Were you aware of AI-based recruitment tools before using the Resume Analyzer system?

a) Strongly Agree ☐ d) Disagree ☐
b) Agree ☐ e) Strongly Disagree ☐
c) Neutral ☐

11. The Resume Analyzer system helps in identifying candidate skills accurately.
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
12. The system correctly analyzes the experience mentioned in the resume.
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
13. The similarity score between resume and job description is useful for recruitment.
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
14. The AI-based Resume Analyzer improves recruitment efficiency.
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
15. The system reduces manual effort in screening resumes.
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
16. What benefits do you see in using AI-based recruitment systems?
- | | | | |
|--------------------------------|-----------------------|-------------------------|-----------------------|
| a) Faster candidate screening | ☐ | c) Reduced human effort | ☐ |
| b) Improved candidate matching | ☐ | d) Better recruitment | ☐ |
17. Do you agree that AI can improve recruitment decision-making?
- | | | | |
|-------------------|-----------------------|----------------------|-----------------------|
| a) Strongly Agree | ☐ | d) Disagree | ☐ |
| b) Agree | ☐ | e) Strongly Disagree | ☐ |
| c) Neutral | ☐ | | |
18. Your level of satisfaction with the Resume Analyzer system?
- | | | | |
|---------------------|-----------------------|------------------------|-----------------------|
| a) Highly Satisfied | ☐ | d) Dissatisfied | ☐ |
| b) Satisfied | ☐ | e) Highly Dissatisfied | ☐ |
| c) Neutral | ☐ | | |
19. Do you feel the system provides accurate candidate ranking?
- | | | | |
|-------------------|-----------------------|-------------|-----------------------|
| a) Strongly Agree | ☐ | c) Neutral | ☐ |
| b) Agree | ☐ | d) Disagree | ☐ |
20. How would you rate your overall experience with the Resume Analyzer system?
- | | | | |
|--------------|-----------------------|------------|-----------------------|
| a) Excellent | ☐ | c) Average | ☐ |
| b) Good | ☐ | d) Poor | ☐ |
21. Suggestions (if any):.....

